LSJ-5000 Instructions

Contents

- Building the sensor (-----coloured slides)
- Installing the software (-----coloured slides)
- Using the sensor (-----coloured slides)

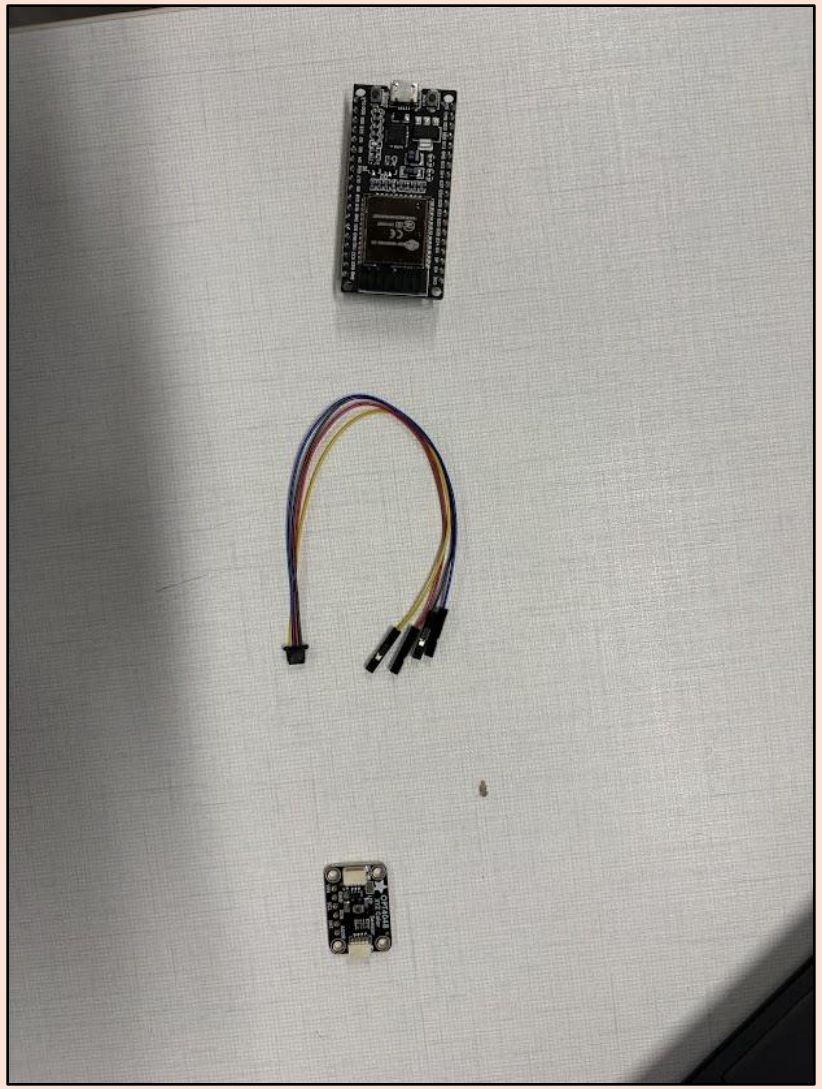
Building the Sensor

What You Need

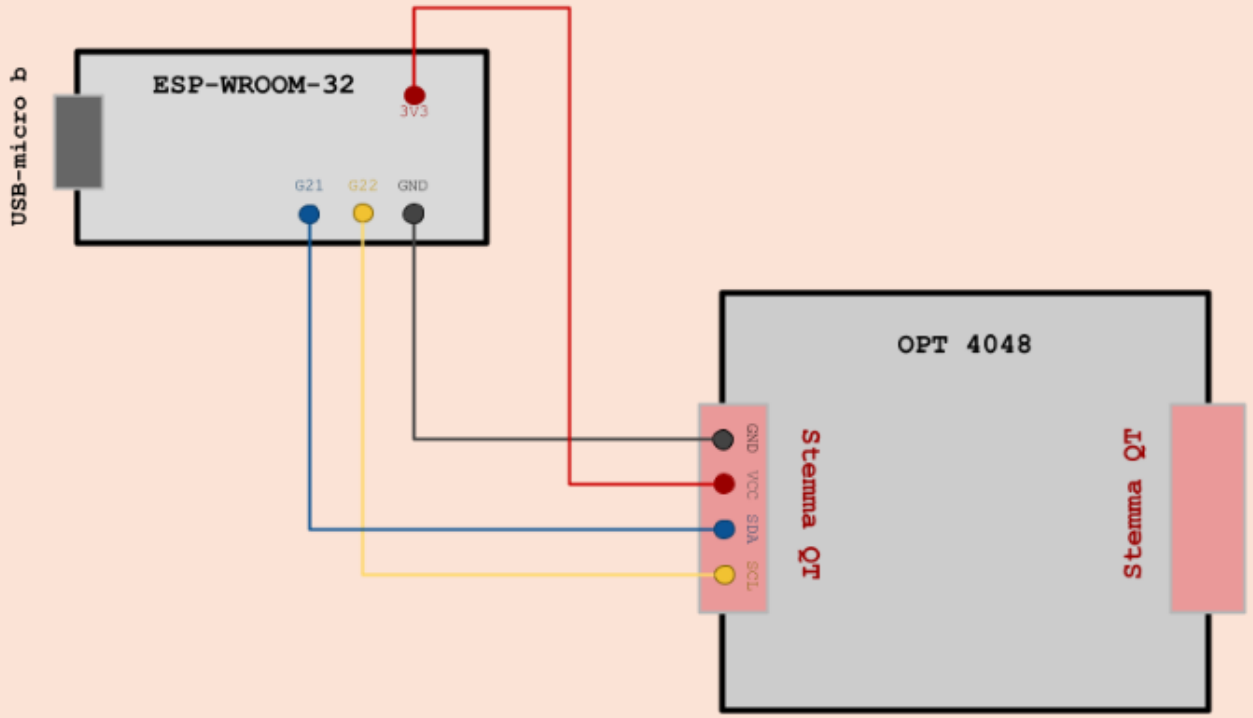
- ESP-WROOM-32 (ESP32 board)
- OPT4048 light sensor
- Jumper wires
- USB cable (for power + programming)

Important:

To get everything working, you only need to connect four wires two for power and two for communication.



Process Diagram Overview

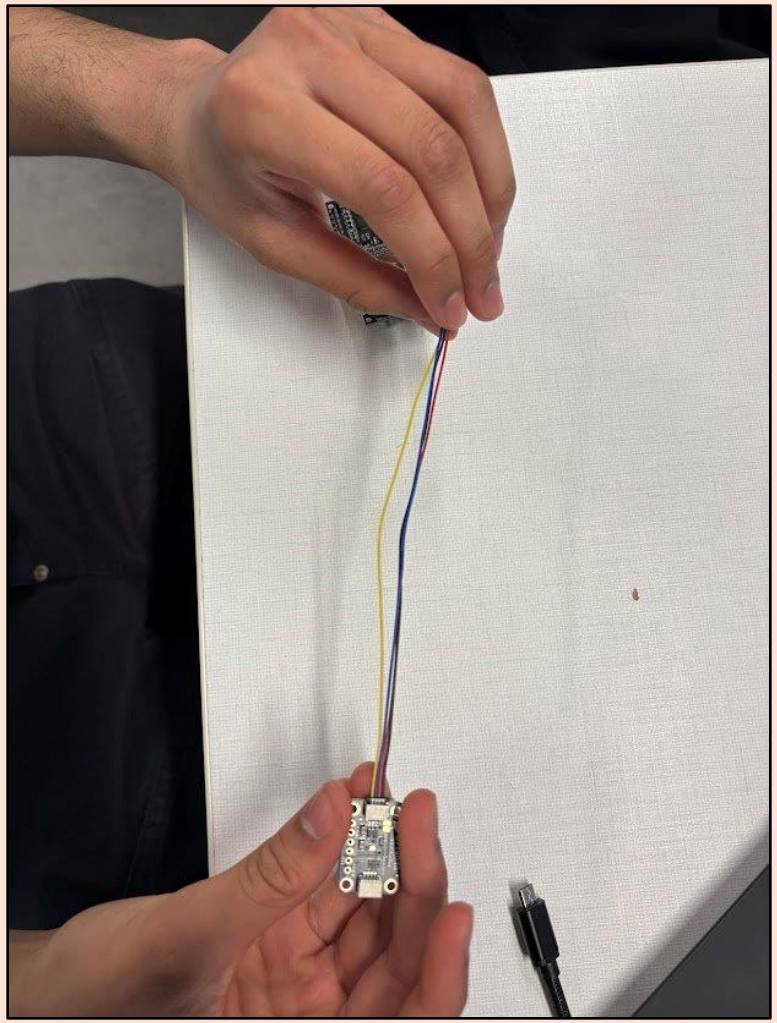


Connect Power

Start by powering the sensor from the ESP32

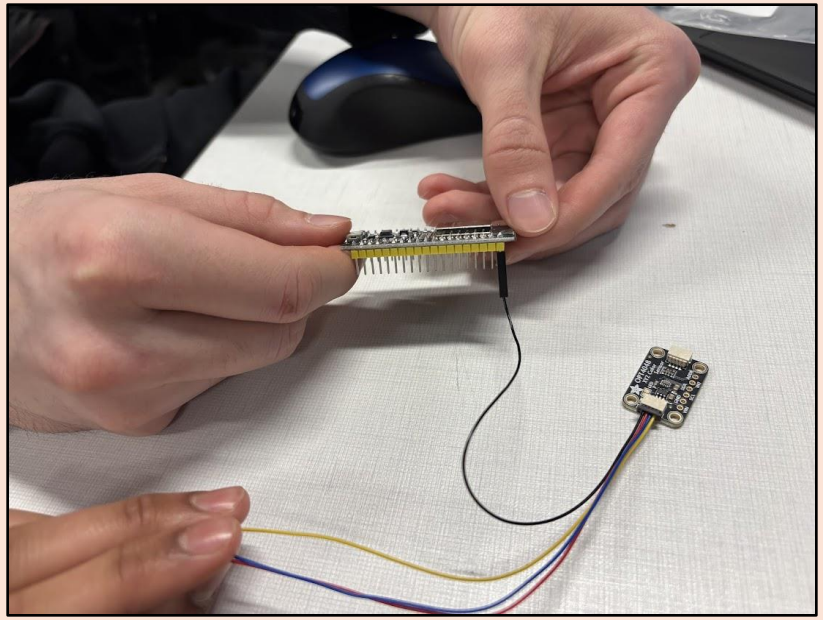
Take your jumper wires and connect them to the sensor.

- Insert the wires firmly into the sensor pins
- Make sure each wire is fully connected and not loose

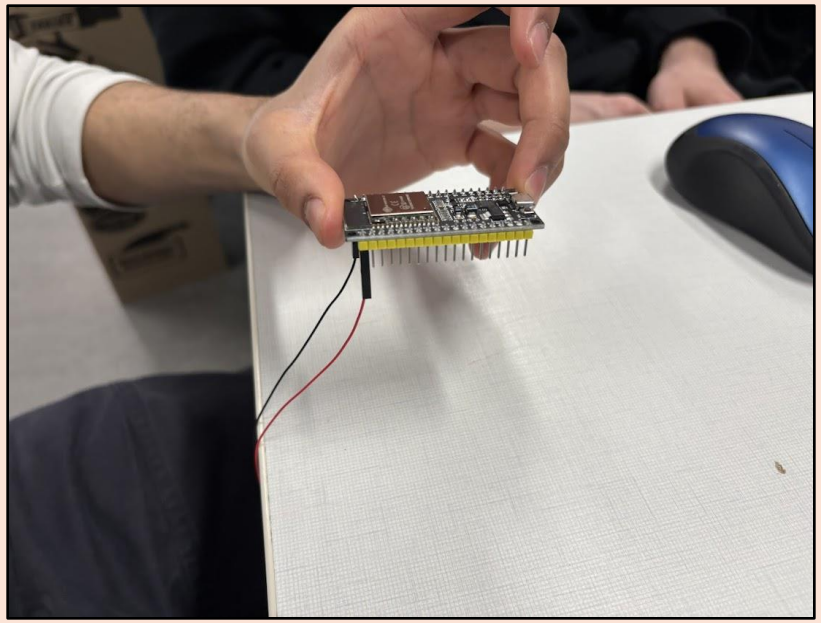


Connect Power

2. Connect GND on the ESP32 to GND on the sensor (black wire)



3. Connect the 3.3V pin on the ESP32 to the VCC pin on the sensor (red wire)

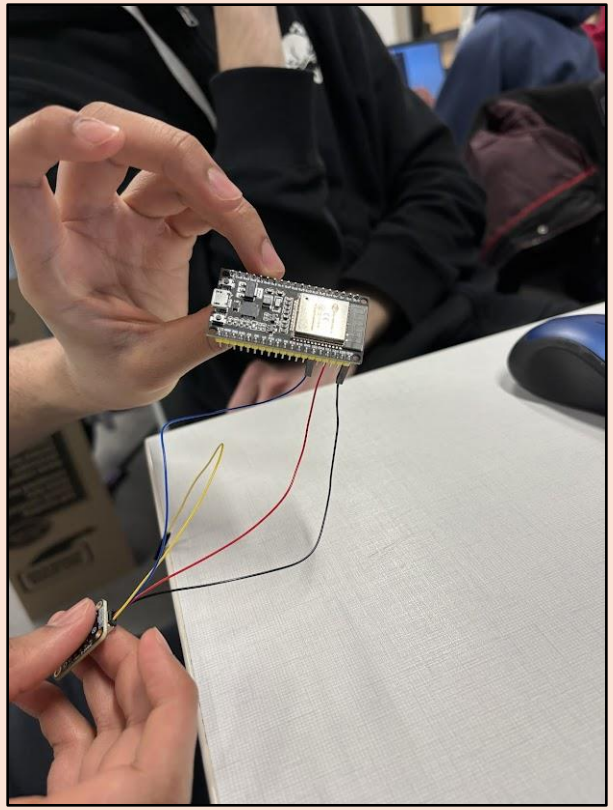


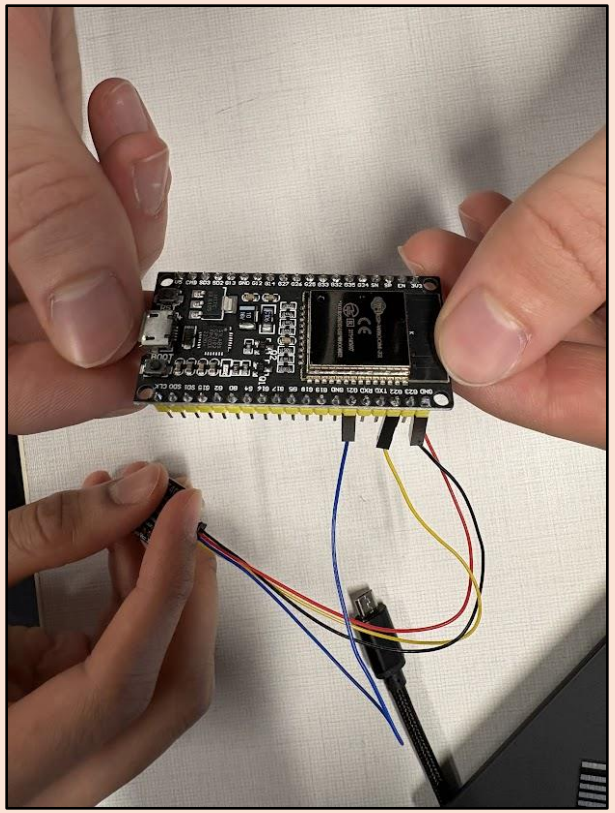
Connect Data

Next, connect the communication wires so the ESP32 can read data from the sensor:

4. Connect pin 21 (SDA) on the ESP32 to SDA on the sensor (blue wire)

5. Connect pin 22 (SCL) on the ESP32 to SCL on the sensor (yellow wire)



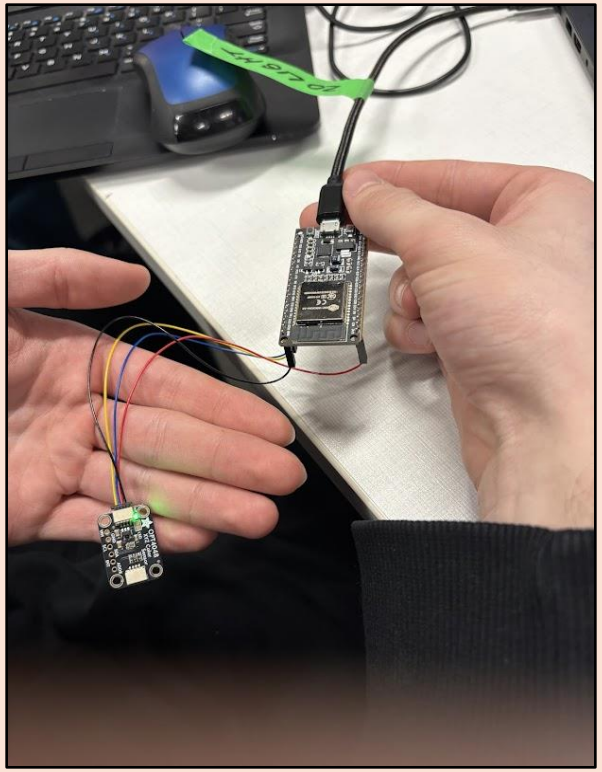


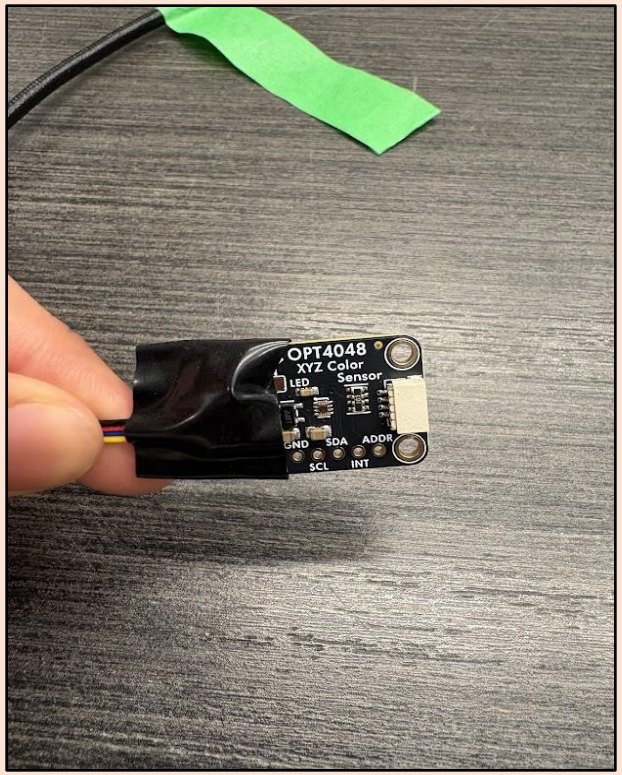
Powering the System

6. Plug the ESP32 into your computer using the USB cable

7. This will power both the ESP32 and the sensor. The sensor should have a green light to indicate the power is successfully connected.

Extra: After connecting the wires, place a small piece of tape over the connection points on the sensor.





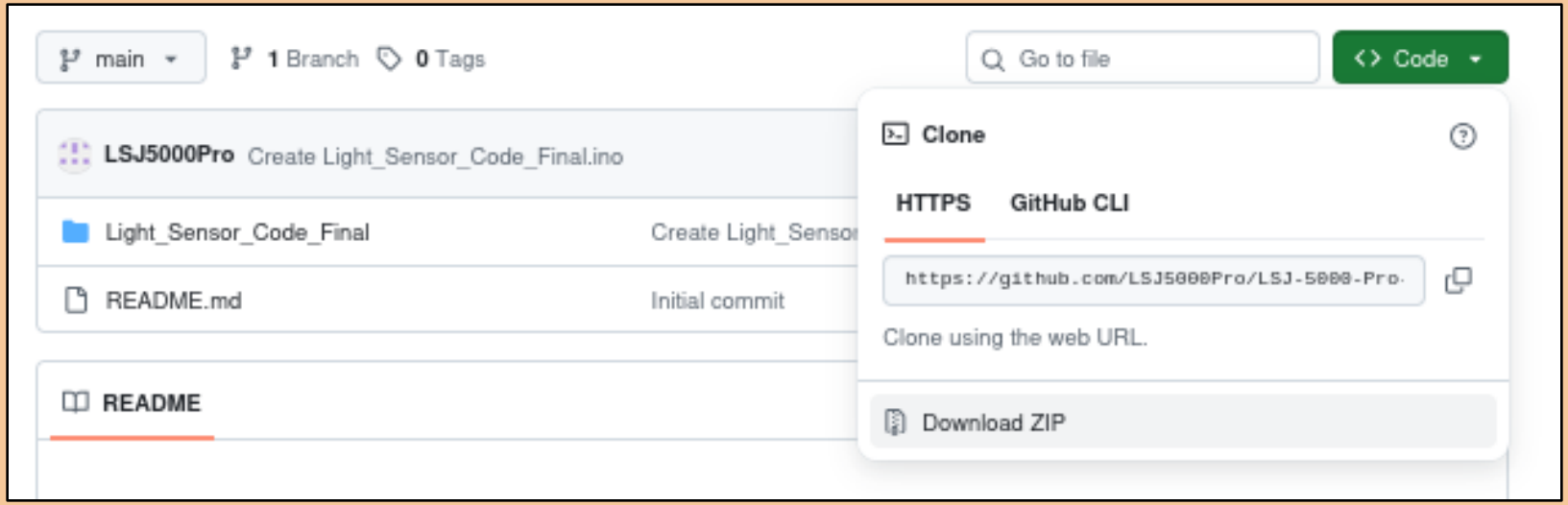
Summary of Pin Mapping

ESP32 Pin	Sensor Pin	Function
3V3	VCC	Power
GND	GND	Ground
G21	SDA	Data (I^2C)
G22	SCL	Clock (I^2C)

Installing the Software

Installing Arduino IDE and Getting the Code

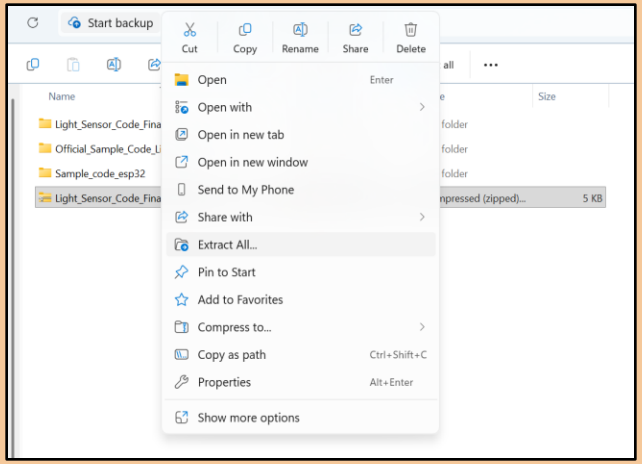
- Install Arduino IDE, and the ESP32 drivers from Espressif, using the following links and instructions
  - https://www.arduino.cc/en/software/
  - https://docs.espressif.com/projects/arduino-esp32/en/latest/installing.html
- Now download the code from Github by clicking on the green "Code" button, then "Download ZIP", it will appear in your downloads folder
  - https://github.com/LSJ5000Pro/LSJ-5000-Pro---Sensor-Code/tree/main

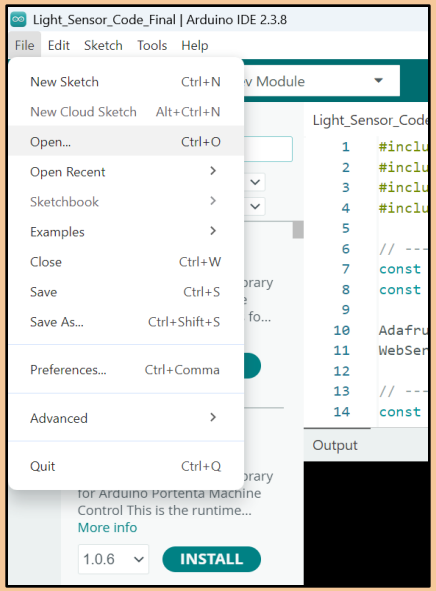


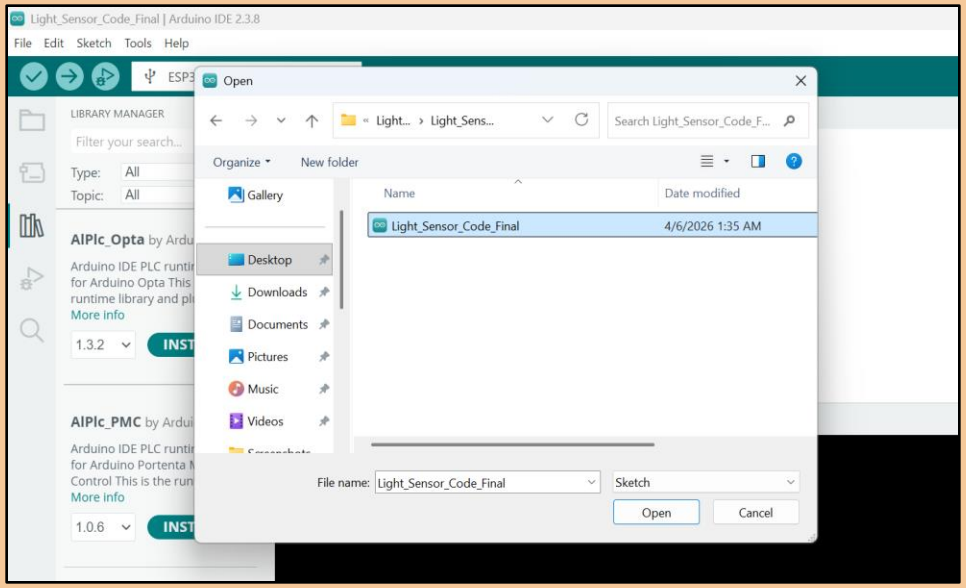
Opening the Code with Arduino IDE

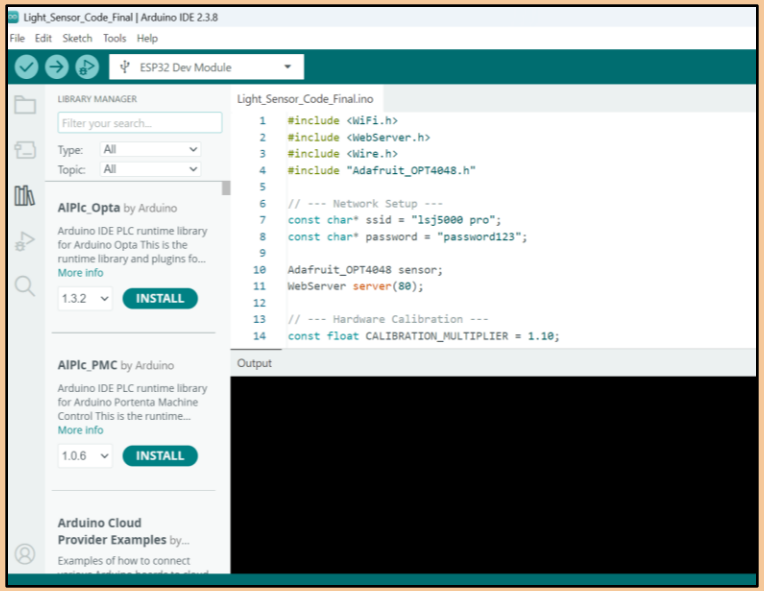
First unzip the file containing the code

- Now Open it in Arduino IDE, <u>you should see the code appear</u>



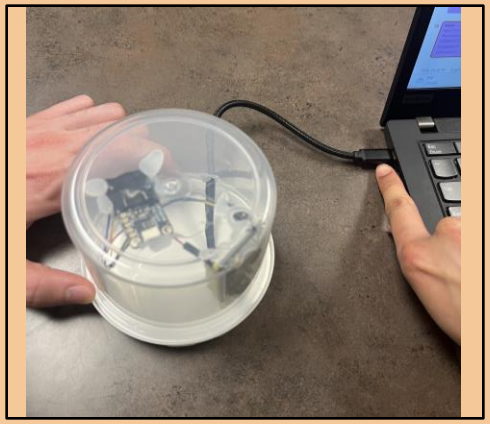




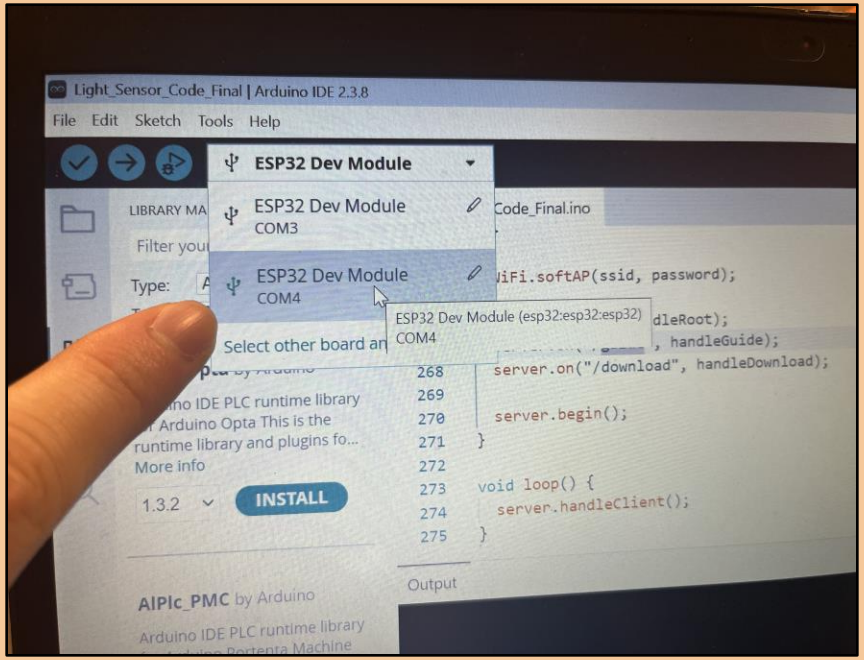


Now Flash the Code to the ESP32

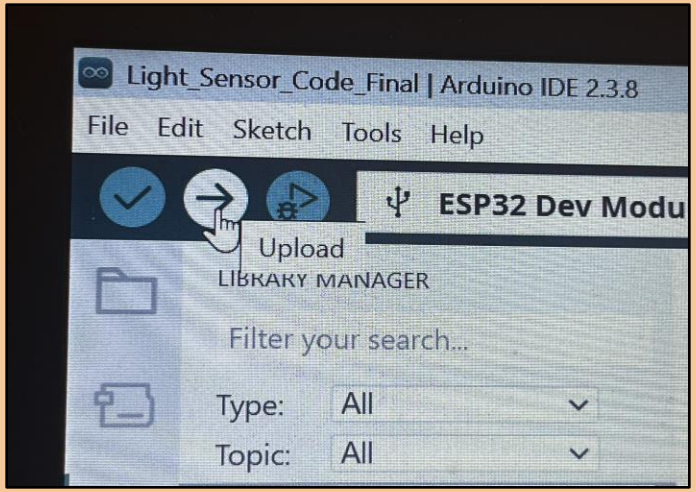
- Plug in the ESP32



- Select it in the menu



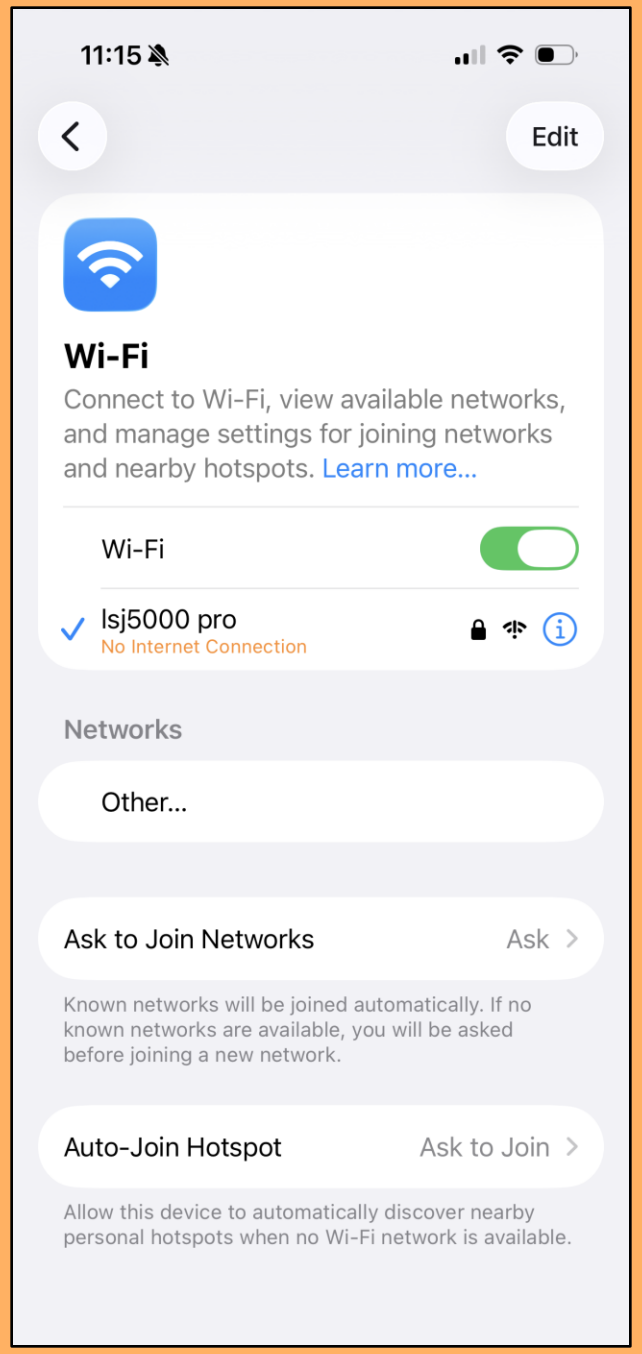
- Upload the code



Using the Sensor

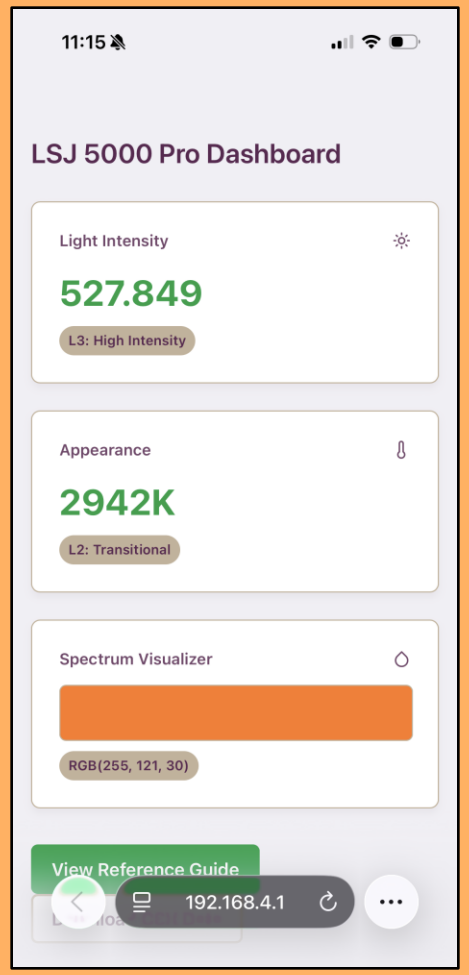
Wi-Fi Network

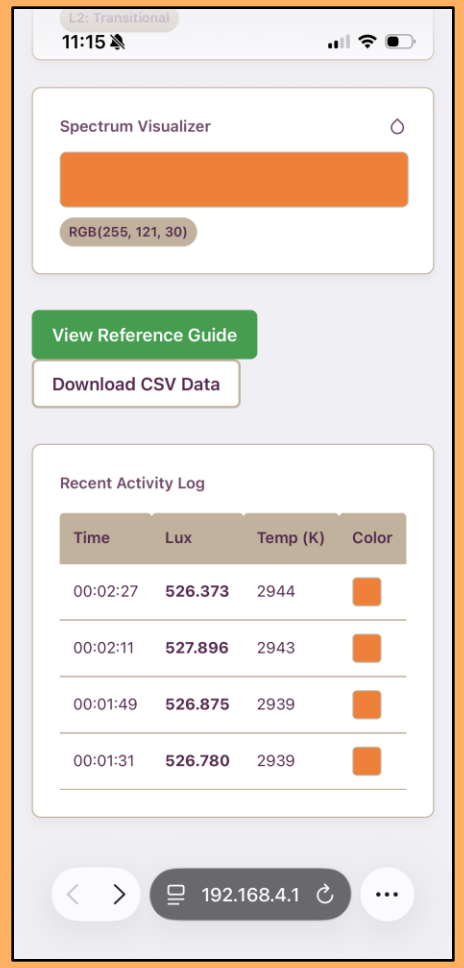
- Once sensor is powered on, go to local Wi-Fi networks and look for Wi-Fi titled "lsj5000 pro"
- Once you spot it, connect to it.
- The password will be "password123"

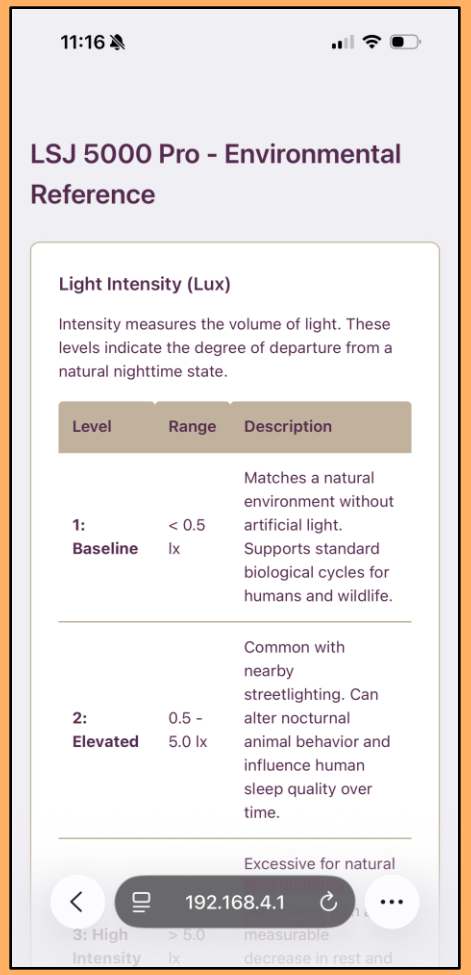


Website

- Go to any browser and type in your 192.168.4.1.
- You should be loaded into a website that looks identical to the one in the picture.
- On this Website you will see the lux value, color temperature, and color that your sensor is measuring.
- Below the live values is a reference guide that explains the values and the link to download the data as a csv.
- At the very bottom is the time logs with the lux, color temperature, and color.







Troubleshooting Table

Issue	Possible Cause	Solution
No data showing	Loose wiring	Check all connections
Sensor not detected	SDA/SCL swapped	Recheck pin connections
Inconsistent readings	Obstruction or movement	Keep sensor steady and unobstructed
Cannot connect to device	Wi-Fi not enabled	Restart device and reconnect